Emanuel André Medina Araujo

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Education

Technical University of Munich — Oct. 2026 – Present
Master of Science in Informatics — *Munich, DE*

Technical University of Munich — Oct. 2020 – Aug. 2026
Bachelor of Science in Computer Science, Application Area in Physics — *Munich, DE*

Technical University of Munich — Oct. 2024 – Sep. 2025
Parallel studies in Physics (B.Sc. programme, not completed) — *Munich, DE*

Herzog-Christoph-Gymnasium — Sep. 2012 – Sep. 2020
Abitur, Awarded MINT-EC certificate with distinction — *Beilstein, DE*

Experience

Working Student Software Engineer — Nov. 2024 – Present
Vector Informatik GmbH - VSceneCreator (3D OPENDrive Editor) — *Munich, DE*

- Helped ship the 0.1 to 0.4 alpha versions of the VSceneCreator (Unity/C#) by suggesting, designing and implementing features, addressing bugs, and creating tests through close collaboration with Product Management.
- Architected a read-only "Explore Mode" with a hierarchical overview tree and dynamic tooltip system, segregating access to editing features resulting in a preview tool for licensed users and an improved UX for free users.
- Improved edit mode switching performance by restructuring the base model to enable lazy loading for non-essential and non-visible objects resulting in up to 70% faster switch times.
- Implemented support for single-sided lanes, enabling users to load OPENDrive files with single-sided lanes by automatically converting them to normal lanes resulting in prevention of broken files.

Working Student Software Engineer — Mar. 2021 – Oct. 2024
Vector Informatik GmbH - DYNA4 (Simulation environment for virtual driving tests) — *Munich, DE*

- Developed features, fixed critical bugs, and enhanced UI using Java and Eclipse RCP, contributing to 5 major product releases (5.0 to 9.0) in direct collaboration with the Product Management.
- Engineered a custom view to assign aliases for trace signals with a new file format, a custom persistence layer and reducing expected user knowledge with errors and warnings via built-in validation and quickfixes.
- Enhanced usability by completely reworking the trace signal view, leading to a more modern and efficient UX.
- Reduced manual QA effort and accelerated development cycles by expanding automated test suites using SWTBot and JUnit in addition to enhancing the CI pipeline to link Jenkins results with JIRA tickets.

Projects

LoveNeverDies | *Legal Loves Tech Hackathon – Bavarian Ministry of Justice Challenge* — Aug. 2026

- Built an end-to-end encrypted estate assistant for bereaved families (React, TypeScript, Supabase) in ten days with a team of lawyers, guiding users through inheritance deadlines via a 141-node legal decision tree.
- Designed a zero-trust architecture in which the server only stores ciphertext: hybrid post-quantum key exchange and signatures (ML-KEM + X25519, ML-DSA + Ed25519), AES-256-GCM, and a Shamir-secret-shared vault.
- Verified row-level security against real Postgres migrations with 130 Vitest files and multi-user Playwright E2E tests.

Learning Based Inverse Kinematics | *Bachelor Thesis* — Jun. 2024 – Nov. 2024

- Compared learning-based IK solvers (MLP, LSTM, hybrid, PPO) for 2- and 3-DOF manipulators in PyTorch on a 10k-sample test set; a simple MLP with direct-angle prediction consistently outperformed the more complex models.

Ironman | *2.8km swim, 180km bike, 42.2km run; Overall Time : 14:21:26h* — Oct. 2022 – Aug. 2024

- Completed full Ironman triathlon through disciplined training and a goal-oriented race plan, demonstrating resilience and on-the-fly problem-solving under fatigue and variable race conditions to reach the finish.

Technical Skills

Languages: C#, Java, TypeScript, JavaScript, Python, C/C++, SQL, MATLAB
Frameworks: React, React Native, Unity, PyTorch, Eclipse RCP, Supabase/PostgreSQL, WebRTC
Developer Skills: Git, Linux, GitHub Actions, Jenkins, Playwright, Vitest, Rider, VS Code, VIM Motions, Jira